

Topology General Exam

January 10, 2022

Instructions: This is a four hour exam. Your solutions should be legible and clearly organized, written in complete sentences in good mathematical style. All work should be your own-no outside sources are permitted-using methods and results from the first year topology course topics. Each problem is worth the same number of points.

Problem 1

Let (X, A) be a pair of topological spaces, and consider the union $X \cup CA$ of X with the cone on A . Here we think of $CA = A \times [0, 1] / \{(a, 1), a \in A\}$, and in $X \cup CA$ we identify $(a, 0) \in CA$ with $a \in A \subset X$.

- a) Prove that X is a retract of $X \cup CA$ if and only if A is contractible in X , i.e., there is a homotopy from the inclusion $A \hookrightarrow X$ to a constant map.
- b) Prove that $(X \cup CA) / X$ is homeomorphic to the suspension SA .

Problem 2

Let $A = \{(0, 0, t) \mid -2 \leq t \leq 2\} \subseteq \mathbb{R}^3$ and $B = \{x \in \mathbb{R}^3 \mid |x| = 2\} \subseteq \mathbb{R}^3$.

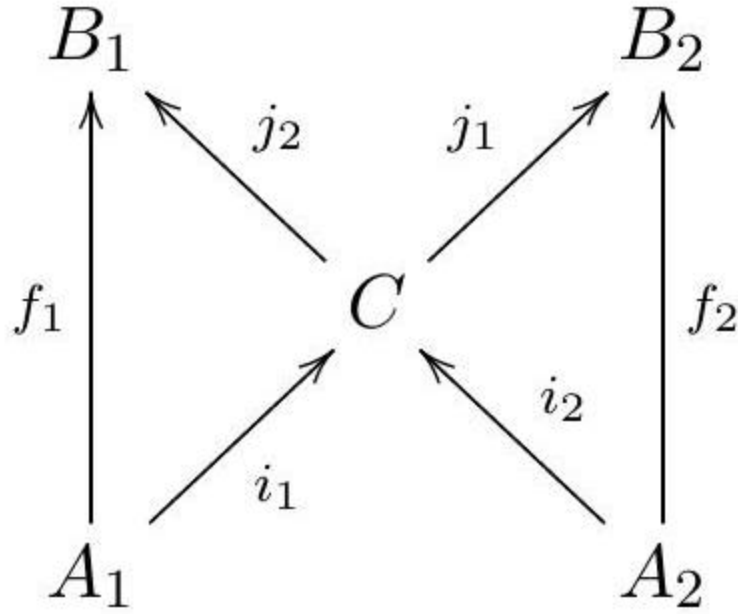
- a) Compute $\pi_1(A \cup B)$.
- b) Compute the homology groups of $A \cup B$.

Problem 3

What are all the connected covering spaces of the 2-dimensional torus T^2 , up to equivalence? Justify your answer.

Problem 4

Consider a commutative diagram of abelian groups



The group C is in the center. Maps i_1 from A_1 and i_2 from A_2 point inward to C . Maps j_2 from C to B_1 and j_1 from C to B_2 point outward. The vertical side maps are f_1 from A_1 to B_1 and f_2 from A_2 to B_2 . The two triangles commute: j_2 composed with i_1 is f_1 , and j_1 composed with i_2 is f_2 .

such that $j_1 i_1 = 0 = j_2 i_2$. Prove that if f_1 and f_2 are isomorphisms and the sequence

$$A_2 \xrightarrow{i_2} C \xrightarrow{j_2} B_1$$

is exact, then the maps

$$i_1 + i_2 : A_1 \oplus A_2 \rightarrow C \text{ and } (j_2, j_1) : C \rightarrow B_1 \oplus B_2$$

are isomorphisms and that the sequence

$$A_1 \xrightarrow{i_1} C \xrightarrow{j_1} B_2$$

is exact.

Problem 5

Let $M \subset \mathbb{R}^3$ be defined by the equations

$$\begin{aligned} x^2 + y^2 - z^2 &= 1 \\ z^2 - xy &= 0 \end{aligned}$$

Prove that M is a smooth manifold and find its dimension.

Problem 6

Let M be a smooth compact manifold of dimension n , and $f : M \rightarrow \mathbb{R}^{n+1} - \{0\}$ a smooth map. Prove that there exists a line through the origin in $\mathbb{R}^{n+1}$ that intersects $f(M)$ in only finitely many points.

Problem 7

Prove that if M is any smooth manifold, then the tangent bundle TM is an orientable manifold.

Problem 8

Prove that any continuous map $f : S^n \rightarrow S^n$ with $\deg(f) \neq (-1)^{n+1}$ has a fixed point. As a suggestion, you might consider proving the contrapositive.